Exploratory Data Analysis (EDA) on Titanic Dataset

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Tools Used: Python, Pandas, Matplotlib, Seaborn

Objective

The objective of this task is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to extract meaningful insights using statistical summaries and data visualizations. The goal is to identify patterns, relationships, and trends that influence passenger survival.

Dataset Description

The dataset contains information about passengers aboard the Titanic. It includes the following key features:

- **PassengerId – Unique ID of passenger**
- **Survived – Survival status (0 = No, 1 = Yes)**
- **Pclass – Ticket class (1 = First, 2 = Second, 3 = Third)**
- **Name – Passenger name**
- **Sex – Gender**
- **Age – Age of passenger**
- **SibSp – Number of siblings/spouses aboard**
- **Parch – Number of parents/children aboard**
- **Fare – Ticket fare**
- **Embarked – Port of embarkation**

The target variable is Survived.

Data Cleaning

The dataset contained missing values and inconsistencies which were handled as follows:

- **Missing values in the Age column were filled using the median value**
- **Missing values in the Embarked column were filled using the mode**
- **The Cabin column was dropped due to a large number of missing values**
- **Data types and structure were verified using .info() and .describe()**

This ensured the dataset was clean and suitable for analysis.

Exploratory Data Analysis

🔷 A. Univariate Analysis

Univariate analysis focuses on individual variables.

- **Age Distribution (Histogram):**
  Most passengers were between 20–40 years old. The distribution is slightly right-skewed.
- **Fare Distribution (Boxplot):**
  Fare shows significant outliers, indicating some passengers paid very high ticket prices. The distribution is right-skewed.
- **Survival Count:**
  More passengers did not survive compared to those who survived, indicating class imbalance.

🔷 B. Bivariate Analysis

Bivariate analysis examines relationships between two variables.

- **Gender vs Survival (Countplot):**
  Female passengers had a significantly higher survival rate compared to males.
- **Passenger Class vs Survival:**
  Passengers in 1st class had the highest survival rate, while those in 3rd class had the lowest.
- **Age vs Survival:**
  Younger passengers had slightly higher chances of survival, but the relationship is not very strong.

🔷 C. Multivariate Analysis

Multivariate analysis studies interactions among multiple variables.

- **Pairplot:**

- **Shows relationships between numerical features like Age, Fare, and Pclass. Survival patterns are visible across these variables.**
- **Heatmap (Correlation Matrix):**

  - **Fare and Pclass are negatively correlated**
  - **Survival has some correlation with Fare and Pclass**
  - **Most variables show weak correlation, indicating independent influence**

Visualizations & Observations

Each visualization provided important insights:

- **Higher fare passengers had better survival chances**

- **Female passengers were more likely to survive**
- **First-class passengers had priority and higher survival rates**
- **Fare distribution contains extreme outliers**
- **Age does not strongly determine survival but shows mild trends**

Key Findings

- **Gender is a strong predictor of survival (females survived more)**
- **Passenger class significantly influenced survival**
- **Higher fare is associated with higher survival probability**
- **Data contained missing values which were handled effectively**
- **No strong multicollinearity found between features**

Conclusion

Exploratory Data Analysis helped uncover important patterns and relationships in the Titanic dataset. Factors such as gender, passenger class, and fare had a significant impact on survival rates. These insights can be further used to build predictive models and improve decision-making.